Semantic Video Understanding

Object-Level Event Detection System

Final-Project Presentation

Prabal Minotra Sarthak Bhudolia Altamash Hasnain Bhuvan Raju

IIITD

April 22, 2026

Problem Statement, Scope & Users

Problem

- Video streams too large for manual review
- Static detectors cannot handle arbitrary user queries
- Hard-coded compliance rules require code changes per domain

Goal: Type plain English $\Rightarrow$ system detects, tracks, and flags events in live video.

Scope

- COCO-80 + open-vocabulary detection
- Spatial relational rules (e.g. *person on bike without helmet*)
- Attribute-level filtering (colour, brand, material)
- Live webcam and uploaded video

Target Users

- **Safety officers**: PPE compliance
- **Retail analysts**: object search in footage
- **Researchers**: zero-shot video query evaluation

Key Challenge

Bridge *real-time speed* (YOLO) with *semantic flexibility* (CLIP) and *user-defined rule logic* (LLM) in a single system.

Related Work: Baseline Methods

1. YOLO Family (v5/v8)
Single-stage detector, $\leq$5 ms/frame. Fixed 80-class COCO vocab.

- × Cannot detect open-vocabulary objects
- × No semantic understanding

2. CLIP (Radford et al., 2021)
Contrastive image-text pre-training; $\cos(\phi_{img}, \phi_{txt})$ as similarity.

- ✓ Open vocabulary, no retraining
- × No spatial localisation: classifies crops only
- × Sliding-window $= O(k)$ forward passes

3. ByteTrack
Multi-object tracking via Kalman filter + two-pool IoU assignment.

- ✓ No Re-ID checkpoint needed
- × Quality degrades at low FPS ($<$5)

4. YOLO-World (Cheng et al., 2024)
Open-vocab YOLO; text-conditioned bounding box detection.

- ✓ Localises any text prompt as a box
- × False-positive rate rises with many prompts
- × 10$\times$ slower than YOLOv8n

Gaps & Methodological Shortcomings

Core Methodological Gap

Using CLIP as a localiser is incorrect. CLIP computes a *global* image-text similarity. Applied via sliding window, it conflates *localisation* and *verification*: background clutter passes the threshold and objects split across window boundaries are missed.

LLM parsing (prior)

- Raw LLM output used directly: no determinism guarantee
- Full spatial rule embedded in `clip_verify`, overloading CLIP's role
- Synonym mismatch ("bike" vs. "bicycle") silently breaks label matching

Hard-coded compliance

- One rule = one `if`-statement; cannot generalise
- Single-frame decisions; no temporal context
- Fixed proximity ignores real-world scene geometry

Our fixes: YOLO-World for localisation · Deterministic normalisation layer · Spatial proximity inference · Online-learning MLP compliance checker.

Dataset and Evaluation Metrics

Datasets

- **Custom live video**: indoor/outdoor webcam; end-to-end evaluation
- **Uploaded clips** (`IMG_7823.MOV`, `IMG_7824.MOV`): handheld objects, tracking, rule ground truth
- **COCO val2017**: YOLOv8 baseline mAP reference
- **CLIP zero-shot**: query similarity distributions

Evaluation Metrics

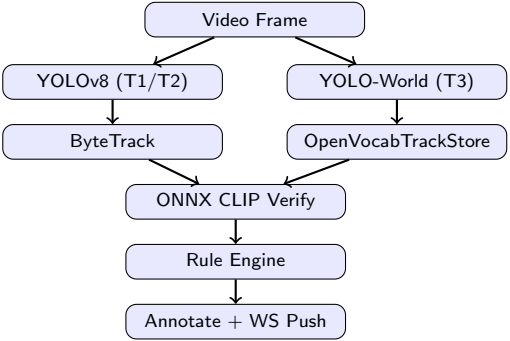
- **mAP @ 0.5 IoU**: YOLOv8 detection accuracy
- **Cosine similarity**: CLIP query relevance (avg 0.231, σ=0.008)
- **Track stability**: ID-switch rate across frames
- **Rule precision / recall**: per annotated clip
- **Frame latency (ms)**: end-to-end per-frame time

Caveat

No public benchmark for NL relational detection; we use manually annotated clips.

System Architecture: Three-Tier Pipeline

- **Tier 1**: YOLOv8 COCO detection (5 ms/frame)
  ByteTrack assigns stable `track_id`
- **Tier 2**: Low-conf YOLO + CLIP re-scoring for weak candidates
- **Tier 3 (new)**: YOLO-World open-vocab for non-COCO
  Strided every $\Delta t = 0.27$ s
  `OpenVocabTrackStore`: candidate $\rightarrow$ confirmed $\rightarrow$ stale
- **ONNX CLIP Verify**: attribute gate ($\cos \geq 0.20$) on primary crops
- **Rule Engine**: IoU + centroid distance spatial logic
- **FastAPI**: MJPEG stream + WebSocket event push



Results: Latency and Detection

Per-frame latency

Stage	Previous	Current
YOLOv8	5 ms	5 ms
CLIP Tier-2	500 ms	<100 ms
YOLO-World	N/A	∼80 ms*
ByteTrack	N/A	1 ms
Rule eval	N/A	0.5 ms

*amortised every $\Delta t = 0.27$ s

CLIP statistics

- Avg. query similarity: 0.231
- Background score: 0.220 vs. object: 0.240
- Variance: 0.008 (stable embeddings)

Rule evaluation observations

- COCO rules (helmet): high precision, robust spatial match
- Non-COCO ("stanley bottle"): YOLO-World localises correctly vs. previous sliding-window miss
- Attribute rules ("grey laptop"): CLIP gate eliminates ∼80% colour mismatches

Positive finding

ByteTrack reduces ID-switches at 2 FPS, enabling per-object compliance history across persons.

Failure Cases and Error Analysis

1. **YOLO-World false positives**
   - Hallucinated boxes for rare prompts
   - Cause: text-conditioned detection over-eager at low conf
   - Fix: `OpenVocabTrackStore` requires $H_c = 2$ hits to confirm
2. **Re-anchoring misfires**
   - Heuristic swaps anchor from person to handheld object
   - Fails for violation rules (e.g. "person using phone while driving")
   - Fix: `is_violation` flag blocks re-anchoring
3. **Small-crop CLIP instability**
   - Crops <40 px give noisy embeddings near 0.20 threshold
   - Cause: ViT patch tokenisation degrades at low resolution
   - Open issue: dynamic threshold based on crop area

Modifications from Previous Submission

Component	Before	After
Compliance rules	Hard-coded `if person & ~helmet`	GPT-4o mini NL parsing: generalises to any domain
Rule normalisation	None; raw LLM output used	Synonym table, re-anchoring, proximity inference
Non-COCO detection	CLIP sliding-window grid (6×8)	YOLO-World: localisation by detector, not CLIP
Sparse detection memory	None	`OpenVocabTrackStore` state machine
Object tracking	2 s gap aggregation	ByteTrack: stable per-object IDs
CLIP inference	Per-crop PyTorch loop $O(k)$	Batched ONNX-Runtime (-80% latency)
Interface	CLI only	FastAPI + React live dashboard

Rationale for Architectural Decisions

YOLO-World instead of CLIP grid

CLIP is a similarity model, not a spatial localiser. The sliding-window forced CLIP to propose regions: a task it was not designed for. YOLO-World correctly separates localisation (detector) from verification (CLIP).

Normalisation layer after LLM parse

LLMs produce non-deterministic outputs. The normalisation layer acts as type-safe schema enforcement: the rule engine never sees vocabulary ambiguity or LLM hallucinations.

OpenVocabTrackStore state machine

YOLO-World runs at reduced stride. Without memory, objects appear/disappear every stride. The store provides temporal persistence: confirmed tracks remain visible for continuous rule evaluation.

ByteTrack over gap aggregation

Gap aggregation cannot dihstinguish two persons sharing a label. ByteTrack assigns per-object IDs enabling "person A has a helmet, person B does not" differentiation.

Next Steps and Individual Contributions

Contributions So Far

Member	Work
Prabal Minotra	YOLOv8Pipeline,ONNX CLIP encoder
Sarthak Bhudolia	RuleEngine,ByteTrack,FastAPI server
Altamash Hasnain	YOLO-World, normalisation,open-vocab memory
Bhuvan Raju	GPT parser,React UI

Open problem

No public benchmark for NL relational event detection: building one is itself a research contribution.